Summer Internship (II) Presentation

on

Smart Irrigation and Monitoring System using ESP32 and RTL/SoC Design

Department of Electronics and Communication
National Institute of Technology, Raipur

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Introduction

- Manual or fixed-schedule irrigation wastes water and stresses crops.
- Another common problem arises when people travel out of town for a few days, as ensuring that their plants receive adequate water during their absence becomes a challenge.
- **Project:** a low-cost smart irrigation controller built in two phases
- **Phase I:** ESP32 IoT prototype with soil, weather and tank sensing, a relay-driven pump and a Blynk cloud dashboard
- **Phase II:** the same algorithm as Verilog RTL, verified, synthesized and implemented as a GDSII layout with OpenLane 2

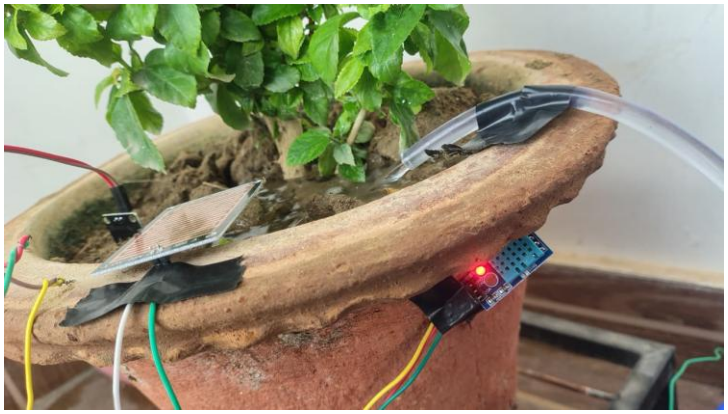


Figure 1: Plant pot, rain sensor and irrigation tube

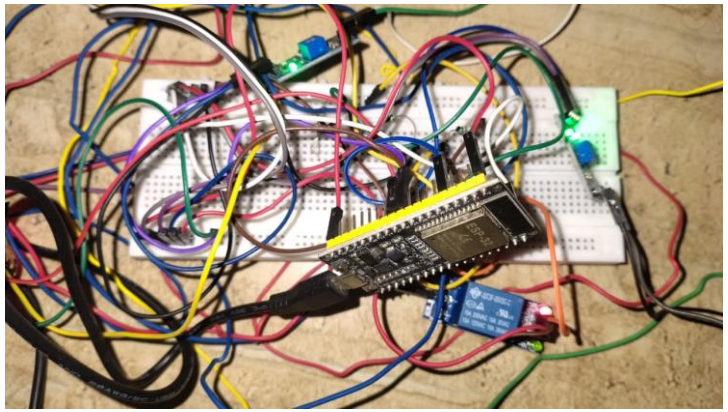


Figure 2: ESP32 prototype wiring

Objectives & Motivation

Objectives

- Automatic irrigation controller on the ESP32
- Cloud monitoring and manual override through Blynk
- Modular Verilog RTL of the same algorithm
- Verify, synthesize and implement it from RTL to GDSII

Motivation

- Agriculture & Plantation is a major consumer of fresh water
- Sensor-driven control applies water only when needed
- Re-targeting the algorithm to hardware shows a path from an IoT prototype to a dedicated chip

Organization Profile

Organization	Innovation and Incubation Hub MNNIT Foundation (IIHMF)
Location	MNNIT Allahabad, Prayagraj, Uttar Pradesh
About	Official incubator of MNNIT Allahabad; central innovation platform of the institute [1]
Functions	Incubation, mentoring and funding access for student and faculty-led ventures [1]
Internship	15 June to 14 July 2026 (4 weeks), under Prof. Basant Kumar, Professor, Department of ECE, MNNIT Allahabad
Relevance	Prototype-to-product setting for the IoT and VLSI work in this project

Literature Survey



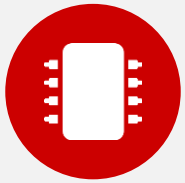
Conventional irrigation

- Fixed timers or manual watering
- No response to soil moisture or rain
- Over- and under-watering



IoT-based irrigation

- Low-cost Wi-Fi microcontroller (ESP32 [2])
- Soil sensor and relay-driven pump
- Cloud dashboards (Blynk [3])



Gap and proposal

- Most work stops at firmware
- Here: same logic re-implemented as verified Verilog RTL [4]
- Carried on to an ASIC layout

System Architecture

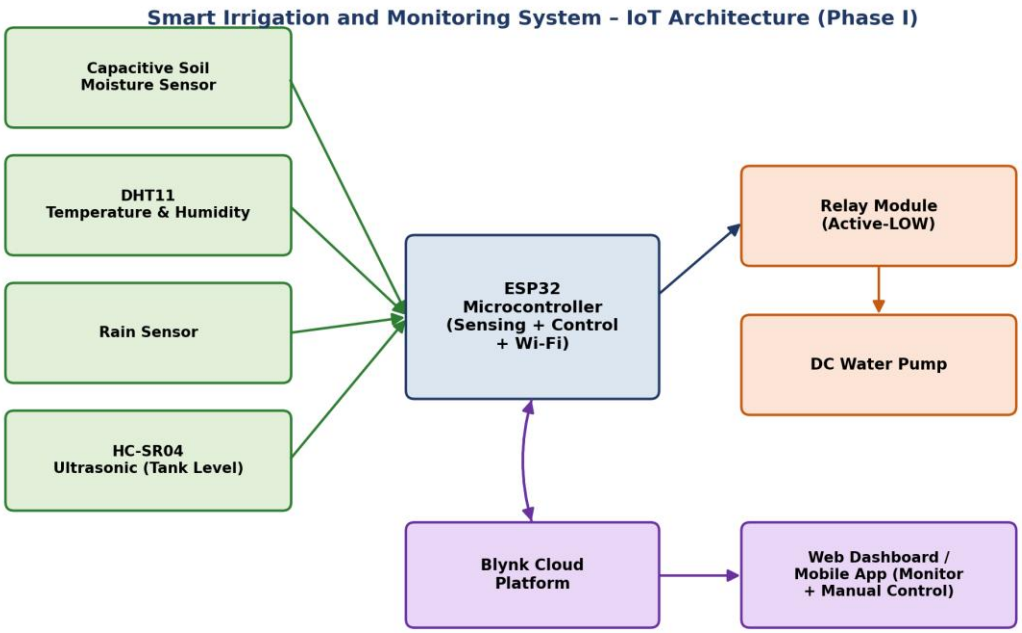
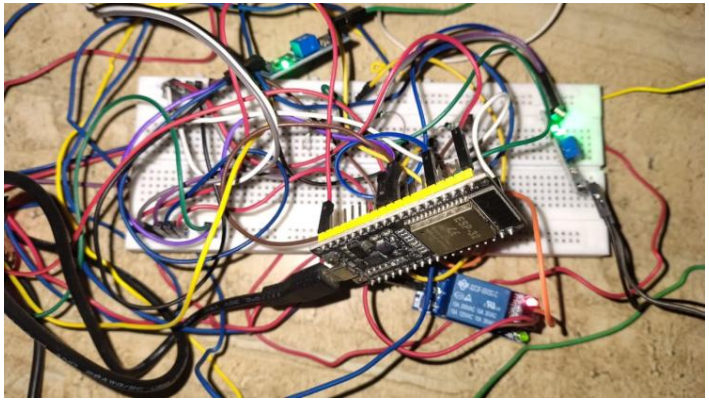


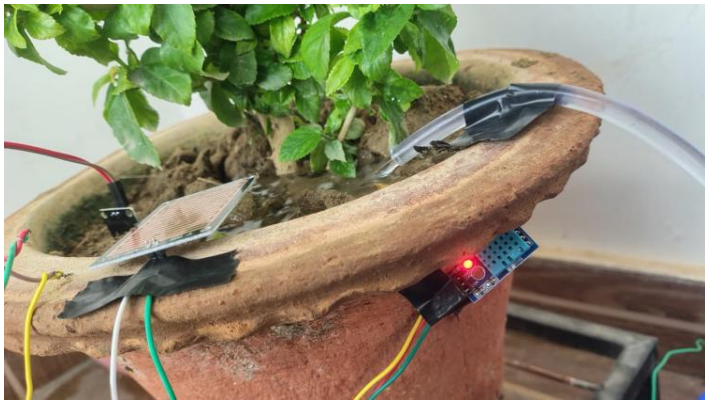
Figure 3: IoT system block diagram

- **Inputs:** soil moisture, DHT11, rain, HC-SR04 tank level
- **Control:** ESP32 thresholds and priorities
- **Output:** relay-driven DC pump
- **Cloud:** Blynk web and mobile, 8 virtual pins
- **Safety:** tank-empty cut-off in every mode

Hardware Prototype



ESP32, relay and wiring



Plant pot with rain sensor



Tank with HC-SR04

ESP32 pin	Connected to
GPIO 34	Soil moisture (ADC)
GPIO 4	DHT11
GPIO 27	Rain sensor
GPIO 26	Relay (pump)
GPIO 18 / 19	HC-SR04 trig / echo

Firmware & Algorithm

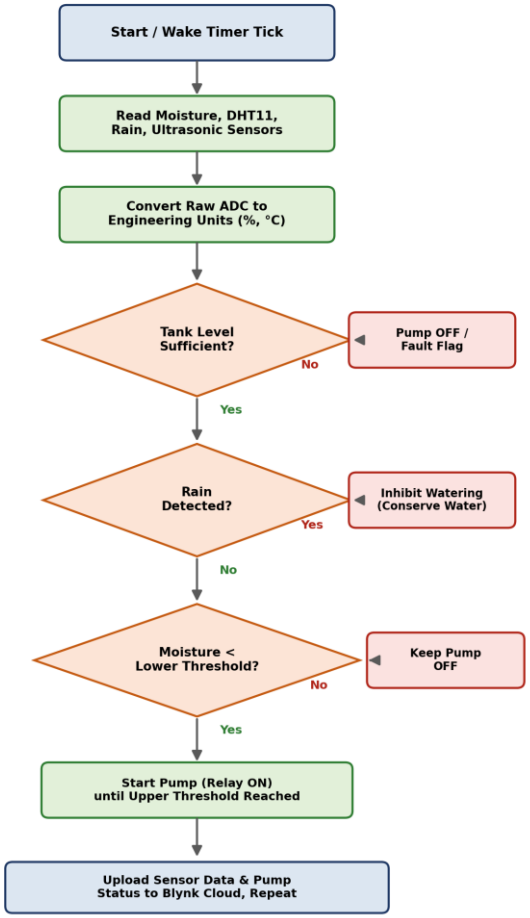
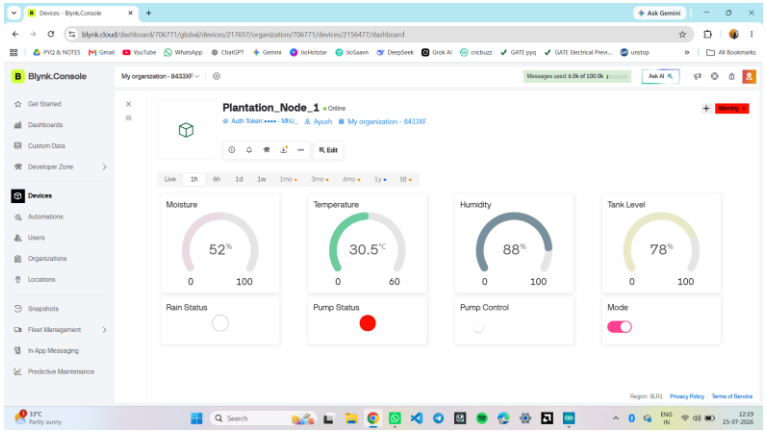


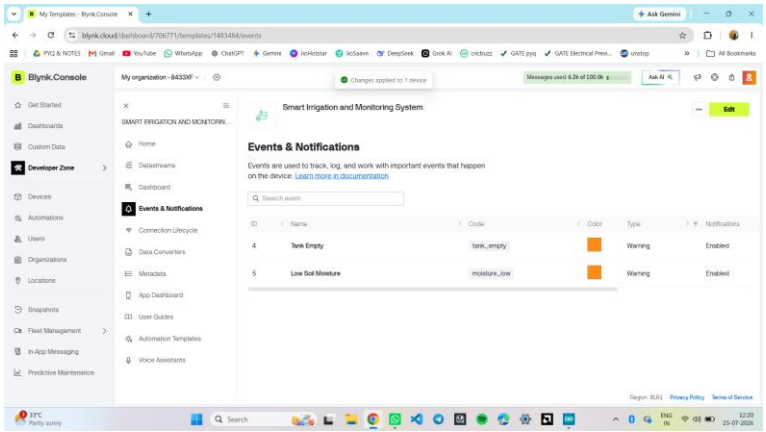
Figure 4: Firmware flowchart

- **Task:** runs on a 2 s timer; moisture averaged over 10 ADC samples
- **Auto mode:** pump ON below 20% moisture, OFF above 60%
- **Rain:** watering inhibited while rain is detected
- **Tank:** pump forced OFF at 10% level or below, with an alert
- **Manual mode:** dashboard switch drives the relay; tank cut-off still applies
- **Cloud:** data to Blynk virtual pins V0 to V7 [3]

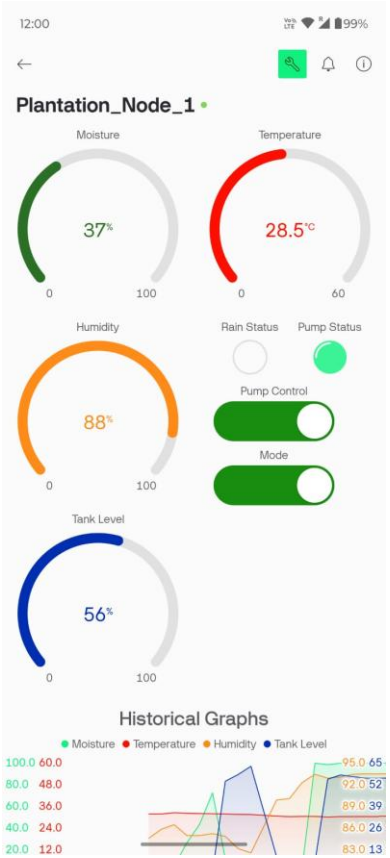
Cloud Monitoring (Blynk)



Web dashboard (manual mode, pump on)



Tank-empty and low-moisture events



Mobile dashboard



History graph

RTL Design of the Controller

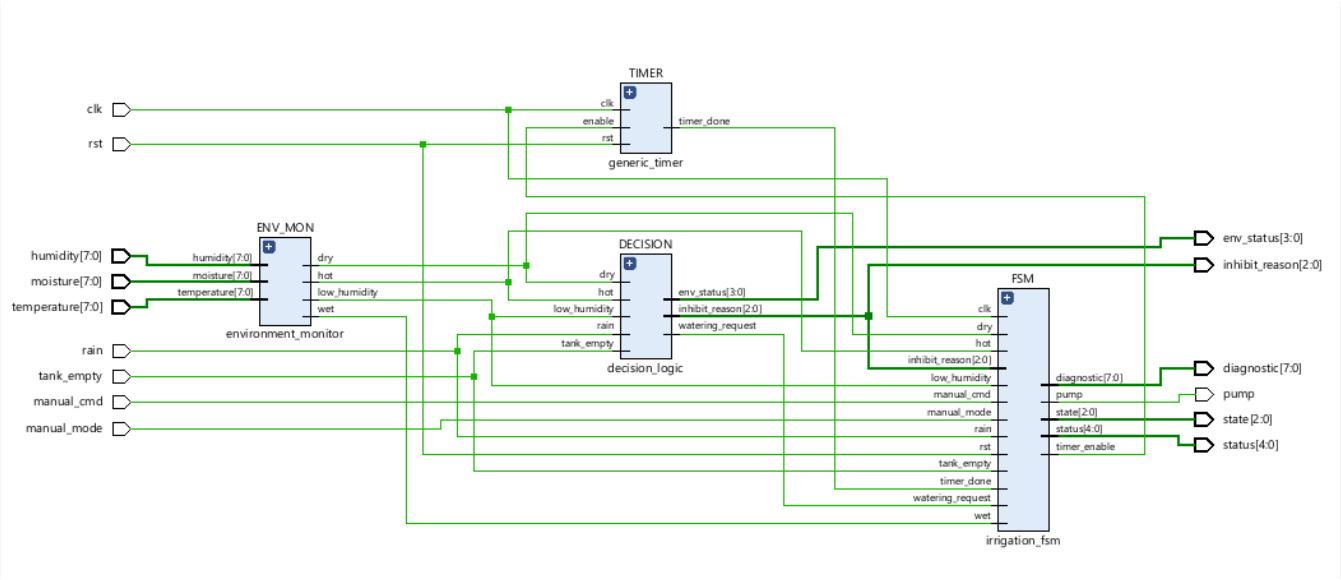


Figure 5: Elaborated RTL schematic of smart_irrigation_top

Environment Monitor

8-bit sensor values to dry, wet, hot, low-humidity flags

Decision Logic

Priority rules give a watering request and inhibit reason

Generic Timer

Parameterized counter sets the watering duration

Irrigation FSM

RESET, IDLE, WATER, FAULT states drive the pump

Irrigation FSM

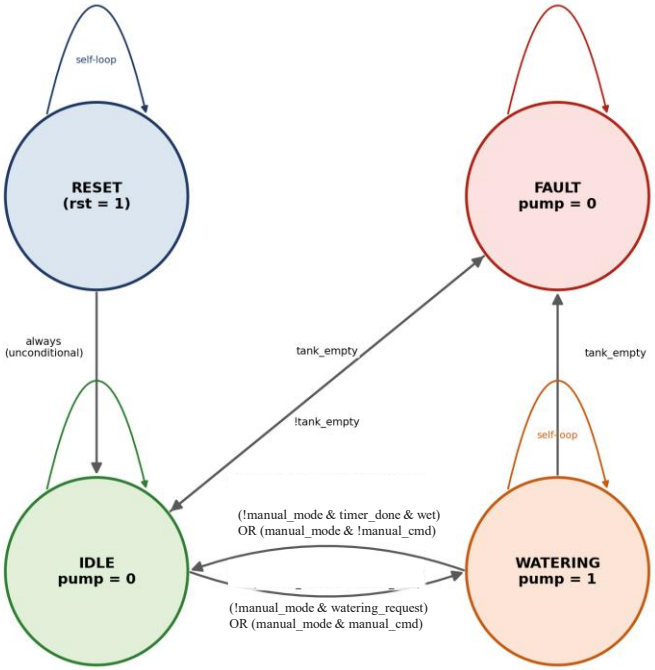


Figure 6: FSM state diagram

Decision priority

Tank empty > Rain > Temperature >

Humidity > Soil not dry

- Tank empty sends IDLE or WATER to FAULT; FAULT returns to IDLE once refilled
- Timer sets the watering duration; moisture above threshold ends it
- Manual mode overrides automation but never the tank-empty cut-off

Functional Verification

#	Scenario	Expected
1	Reset	IDLE, pump off
2	Dry soil	WATER, pump on
3	Timer expiry	IDLE, pump off
4	Rain	Inhibited
5	Tank empty	FAULT, pump off
6	Recovery	IDLE
7a	Manual on	WATER, pump on
7b	Manual off	IDLE, pump off

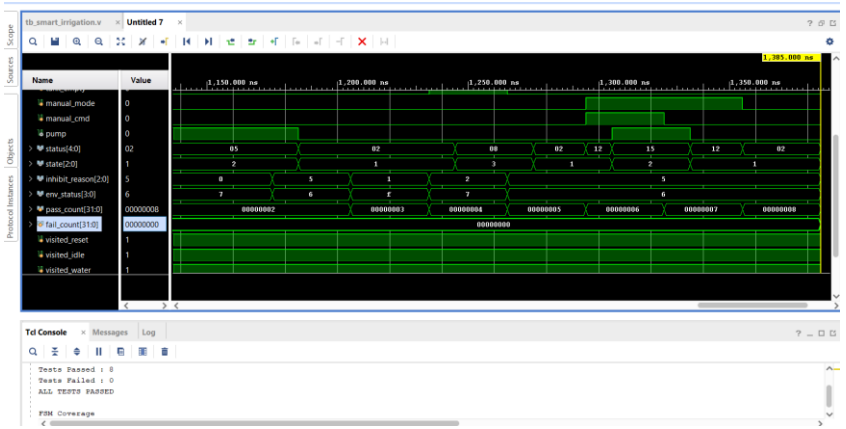


Figure 7: Simulation result

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tests passed

100%

FSM state coverage

Synthesis in Vivado

25

LUTs (0.13%)

12

Flip-flops (0.03%)

53

I/O pins (32.7%)

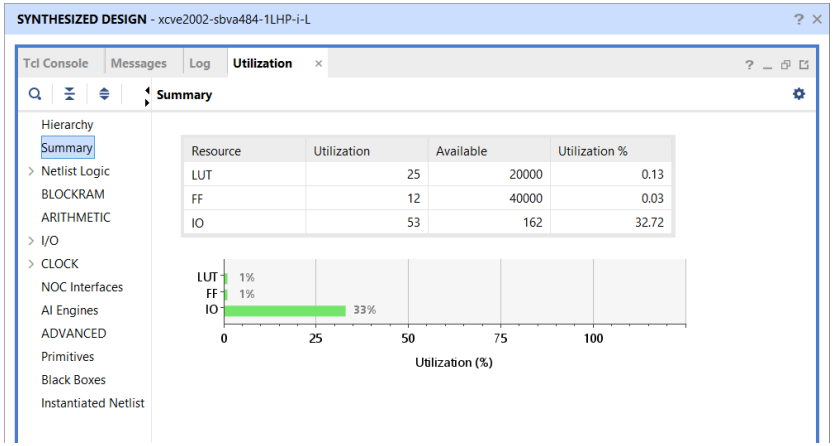


Figure 8: Utilization summary

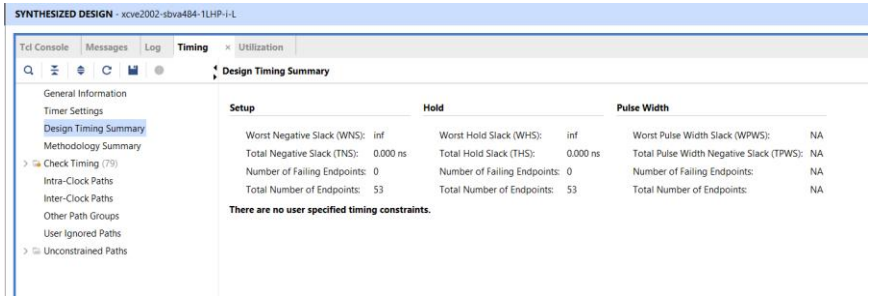


Figure 9: Timing summary (unconstrained)

Target: Versal xcve2002 reference part, AMD Vivado 2025.2 [5]



RTL-to-GDSII with OpenLane 2

1 Synthesis	**2 Floorplan & PDN**	**3 Placement**	**4 Clock tree**
5 Routing	**6 STA, 9 corners**	**7 DRC, LVS, antenna**	**8 GDSII**

Setup

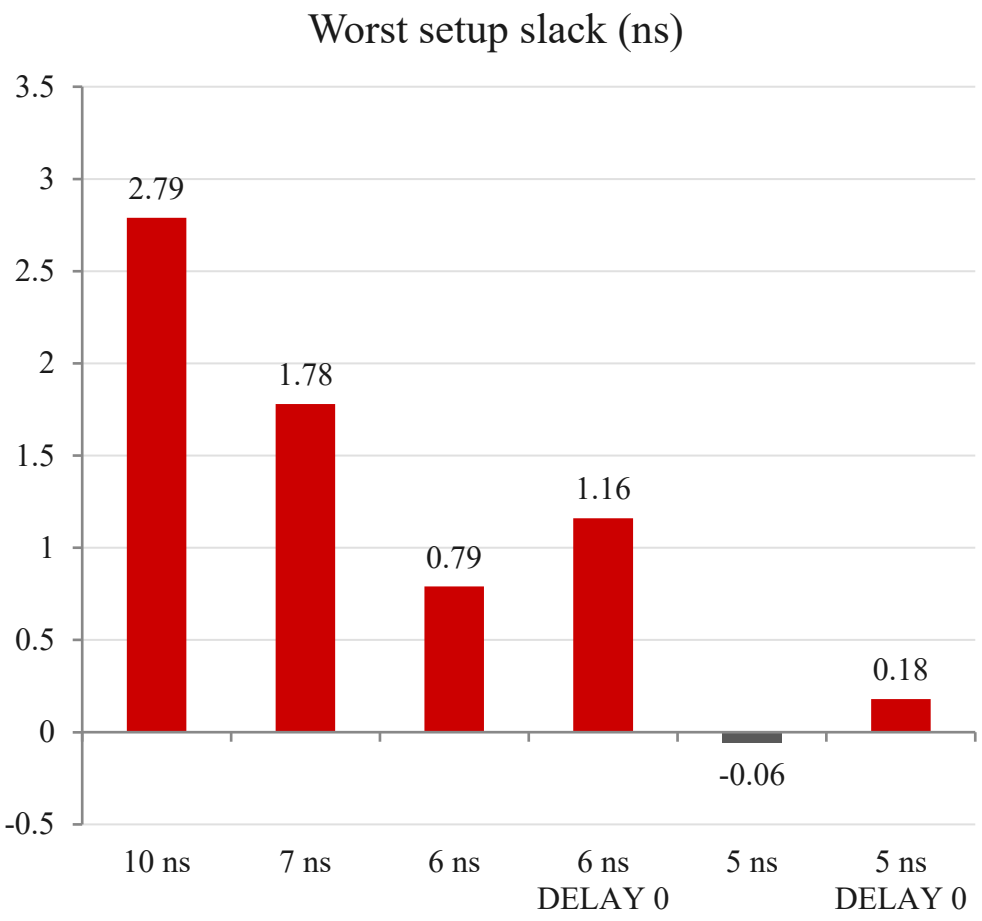
- SKY130A, sky130_fd_sc_hd cells [6][7]
- Yosys / ABC synthesis, OpenROAD place and route [8][9]
- 120 x 120 um die, 5 ns clock
- Generic SDC, 20% I/O delays

Finding: synthesis strategy

- Default AREA 0 left a weak 5-input AOI gate: 3 max-slew violations at the slow corner
- DELAY 0 re-mapped the logic: 0 violations
- Cost: about 5% more area



Clock Sweep & Signoff



Final run (5 ns)	Result
Setup slack	+0.18 ns
Hold slack	+0.119 ns
DRC / LVS	0 / 0
Antenna	0
Slew violations	0
Power	391 uW
Std-cell area	1,378 um2

Final Layout & Gate-Level Check

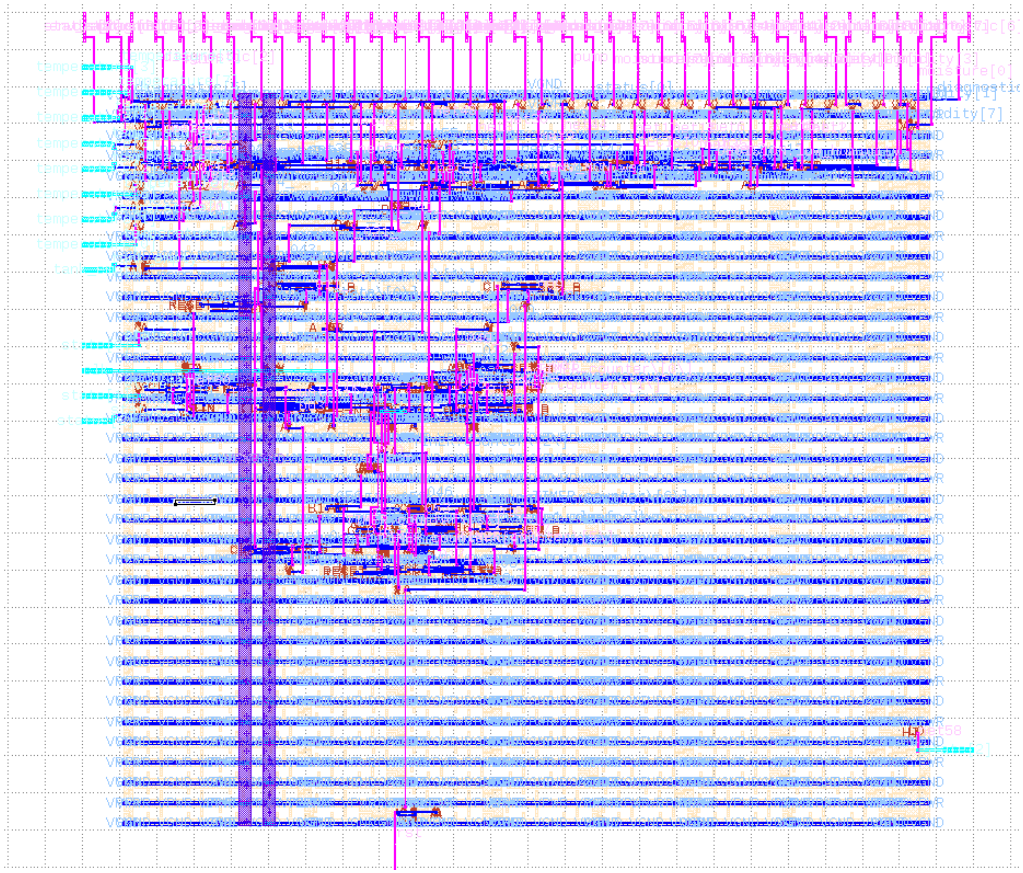


Figure 10: Final routed layout (120 x 120 um)

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gate-level tests pass on the post-layout netlist

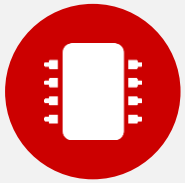
0

DRC, LVS and antenna violations

Race found

reset released on a clock edge; fixed with a 1 ns offset

Key Learnings & Skills



Technical Knowledge

- ESP32 firmware, sensors and Blynk
- Verilog RTL and FSM design
- Self-checking verification
- Vivado and OpenLane 2 flows



Practical & Analytical

- Comparing options with data (clock sweep, strategies)
- Reading STA and tool reports
- Debugging a race and a slew violation



Professional

- Technical documentation
- Treating warnings and failures as feedback
- Stating assumptions and limits of results

Results & Discussion

Aspect	ESP32 IoT (Phase I)	Verilog RTL (Phase II)
Execution	Sequential software	Parallel synchronous hardware
Monitoring	Blynk cloud, web and mobile	Diagnostic output buses
Verification	Live testing on plant and tank	8 self-checking tests, 100% coverage, gate-level sim
Implementation	ESP32 board	25 LUTs, 12 FFs; SKY130 layout, 1,378 um2
Next step	Standalone product	Reusable SoC IP block

The prototype fixed the thresholds; the RTL flow turned them into verified, implementable hardware.

Conclusion & Future Scope

Conclusion

- ESP32 prototype validated with automatic and manual irrigation
- RTL passes 8 of 8 tests with full state coverage
- RTL-to-GDSII at 5 ns: clean DRC, LVS, antenna
- Gate-level simulation matches the RTL

Future Scope

- AXI-style interface for SoC integration
- Digital I2C/SPI sensors, ML-based irrigation prediction
- Solar-powered field deployment
- FPGA board test, custom SDC and SDF timing simulation

References

[1] "MNNIT Incubator Ties Up With Wadhwani Foundation To Start Liftoff Acceleration Program," Wadhwani Foundation. [Online]. Available: https://wadhwanifoundation.org/?p=43622

[2] Espressif Systems, "ESP32 Series Datasheet," Shanghai, China. [Online]. Available: espressif.com

[3] Blynk Inc., "Blynk IoT Platform Documentation." [Online]. Available: https://docs.blynk.io

[4] IEEE Standard for Verilog Hardware Description Language, IEEE Std 1364-2005, 2006.

[5] AMD, "Vivado Design Suite User Guide: Synthesis (UG901)," 2025.

[6] "OpenLane 2 Documentation." [Online]. Available: https://openlane2.readthedocs.io

[7] SkyWater Technology, "SkyWater Open Source PDK (SKY130) Documentation." [Online]. Available: https://skywater-pdk.readthedocs.io

[8] C. Wolf, "Yosys Open SYnthesis Suite." [Online]. Available: https://yosyshq.net/yosys/

[9] T. Ajayi et al., "Toward an open-source digital flow: First learnings from the OpenROAD project," in Proc. 56th ACM/IEEE DAC, Las Vegas, NV, USA, Jun. 2019.



Thank You

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